

Usama Ahmed

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EXPERIENCE

Research Assistant – Machine Learning & NLP

Jan 2025 – Present

University of Arizona · Tucson, AZ

- Framed complex research questions in mathematical terms and developed ML solutions to study logical reasoning in LLMs; applied supervised training over synthetic corpora to improve model quality and generalizability.
- Designed and executed **6 model development experiments end-to-end** — data collection, exploratory data analysis, corpus synthesis (**30K training examples**), model training, testing, and selection — applying statistical hypothesis testing and ablation studies to validate results.
- Generated **30K+ synthetic formal logic deduction examples** using the **FLD corpus generator** (NeurIPS 2024) — configuring argument types (axioms, theorems, references), negation weights, theorem/reference tree probabilities, and step ranges to produce diverse proof-tree corpora; used generated corpora as supervised training data to study and improve logical reasoning in LLMs.

GenAI Intern – Search & Retrieval

Oct 2025 – Dec 2025

Pi Labs, Inc. · San Francisco, CA

- Developed ML-powered **semantic search and question answering pipelines** integrating BM25, dense retrieval, and query fan-out mechanisms; applied data mining techniques to enhance document recall, precision, and ranking relevance.
- Engineered **synthetic data generation workflows** using LLMs (GPT-4o, Claude 3.5) to create **2.17M diverse query-document pairs**; benchmarked semantic vs. lexical retrieval against the full MS MARCO dataset (**8.8M passages**) to drive product search quality improvements.
- Designed **offline evaluation frameworks**, defining metrics (F1, recall@k) and automating gold-label generation to produce decision-ready insights on system performance and regressions.
- Communicated tradeoff analyses (quality, latency, cost) and findings to non-technical product stakeholders as clear written narratives, **directly influencing pipeline architecture decisions**.

Data Scientist II – Applied AI

Nov 2022 – Dec 2023

Afiniti Software Solutions · Washington, D.C.

- Applied predictive modeling and machine learning on large-scale transactional data (**3M+ call volume**) for major telecom clients (Bouygues Telecom and Telefónica), driving a **\$5M/month uplift** in customer retention.
- Executed the full ML lifecycle — problem framing, exploratory data analysis, feature engineering (**20+ new features** on top of 80+ existing), model training, testing, and deployment — automating Python and SQL pipelines for **400K+ entities**.
- Performed statistical analyses and incrementality measurement against baseline/control policies; improved incremental lift from **17 to 33 cents per dollar**.
- Designed real-time monitoring for model health and business outcome regressions (drift, KPIs, anomalies) using PostgreSQL, Python, and R; delivered **executive-facing dashboards** presenting technical results to non-technical senior leadership.

Data Science Product Manager

Nov 2021 – Aug 2022

CarFirst · Lahore, PK

- Built a **Bayesian probabilistic classification system** for operational decisioning, boosting auction success rate by **20%** and predicting a 3% revenue increase.
- Performed ad-hoc statistical analyses and data mining on operational data; translated findings into business insights to drive prioritization and automation of the procurement workflow.
- Delivered recurring data visualization and executive-ready reporting to senior leadership, preventing a **PKR 40M loss** through data-driven harm mitigation.

Graduate Teaching Assistant – SQL/NoSQL Databases

Jan 2025 – May 2025

University of Arizona · Tucson, AZ

- Assisted 50+ students mastering SQL and NoSQL databases, covering indexing, query optimization, and data modeling; led **10+ recitation sessions** and **20+ office hours** troubleshooting complex queries and database design challenges.
- Graded and provided feedback on **100+ assignments** evaluating SQL queries, NoSQL schema designs, and ETL processes; mentored students on data consistency models, CAP theorem, and scalability strategies.

PROJECTS

Elliott Wave Analyzer – ML Pattern Detection & Scoring

[GitHub](#)

Pipeline

- Python · AWS SageMaker · boto3 · S3 · PySpark · XGBoost · Optuna · asyncio · multiprocessing · yfinance · HuggingFace Datasets · Parquet
- Built end-to-end pattern detection pipeline identifying Elliott Wave impulse and corrective structures across **500+ validated patterns / 80+ symbols** (hourly, daily, weekly timeframes) with **100% Elliott Wave rule validation rate** across 21 automated tests.
 - Deployed **XGBoost binary scorer on AWS SageMaker** (managed training jobs + real-time inference endpoint via boto3): engineered **280K synthetic training patterns** using PySpark-based Multivariate Normal modeling with 4-tier Fibonacci constraint filtering; managed S3 data versioning, model artifacts, and endpoint configuration.
 - Implemented **async multiprocessing scanning pipeline** with shared memory, vectorized batch pre-scoring (512–2048 batch size), and skip-ahead deduplication; built backtesting framework measuring Sharpe ratio, max drawdown, win rate, and profit factor across equities, ETFs, FX, and crypto.

Weak Labels Pipeline – Auto-Qrels & Training Triples at Scale

[GitHub](#)

- PyTorch · BM25S · BGE Embeddings · Cross-Encoder (ms-marco-MiniLM) · vLLM · SLURM (HPC) · HuggingFace Transformers · Sentence-Transformers · SQLite · asyncio
- Built a **5-stage weak supervision pipeline** generating relevance labels at million-scale: BM25S (top-200) → BGE dense reranking (top-100) → cross-encoder scoring (top-40) → smart diversity sampling → vLLM LLM judge (Llama 3.2 3B, 20 async workers); operationalized on **SLURM HPC** with multi-GPU distributed inference.
 - Reduced weak-label judging latency from **13s/query to ~2s/query** via vLLM batching and async scheduling; produced **665K+ relevance judgments** and **333K training triples** (query, positive, hard negatives) with SQLite caching and checkpoint/resume support.

Synthetic Query Generation Pipeline (RAG / Search Evaluation)

[GitHub](#)

- LangChain · BERTopic · Sentence-Transformers · BM25 (rank-bm25) · llama.cpp · scikit-learn (TF-IDF + cosine) · PyTorch · asyncio · httpx
- Developed topic-aware synthetic query generation pipeline over Wikipedia using **BERTopic topic modeling** + locally served **LLaMA via llama.cpp**; generated 6 query types (informational, exploratory, navigational, comparative, transactional, commercial) per discovered sub-topic.
 - Filtered **1M+ candidate queries** via multi-stage hybrid heuristics — entropy scoring, BM25 passage relevance, TF-IDF near-dedup, Sentence-Transformer semantic similarity, and **MMR-style diversity enforcement**; rejection-sampled to hit per-type targets and published to HuggingFace Hub.

EDUCATION

M.S. Data Science

Jan 2024 – May 2025

University of Arizona · Tucson, AZ · Presidential Scholar · Dean's Honor List 2025

B.S. Economics & Mathematics

Aug 2017 – Jan 2022

Lahore University of Management Sciences · Lahore, PK · Dean's Honor List 2021

TECHNICAL SKILLS

Languages:	Python · SQL · R · Bash
Machine Learning:	XGBoost · Scikit-Learn · Supervised & Unsupervised Learning · Predictive Modeling · A/B Testing · Statistical Hypothesis Testing · Experiment Design · Anomaly Detection · Optuna · SHAP · Feature Engineering
Deep Learning / NLP / GenAI:	PyTorch · TensorFlow · Keras · HuggingFace Transformers · RAG · Semantic Search · LLM Fine-Tuning · Cross-Encoder Reranking · Weak Supervision · Synthetic Data Generation · BM25 · Dense Retrieval · vLLM · llama.cpp
Cloud & MLOps:	AWS SageMaker · boto3 · S3 · SLURM (HPC) · Distributed Inference (vLLM · multi-GPU) · Git/GitHub · asyncio · multiprocessing
Data Engineering:	PySpark · PostgreSQL · MySQL · MongoDB · Parquet · JSONL · ETL Pipelines · HuggingFace Datasets Hub
Visualization & Reporting:	Matplotlib · Seaborn · Plotly · Power BI · ggplot2 · R Markdown · Executive Dashboards